

Problem #1

```
import numpy as np
```

```
#a
```

```
x = np.zeros(12)
```

```
x[4] = 1
```

```
print(x)
```

```
#b
```

```
y = np.random.rand(10)
```

```
sort_y = np.sort(y)
```

```
print("original ", y)
```

```
print("sorted ", sort_y)
```

```
#c
```

```
z = np.linspace(1, 2, 10)
```

```
print(z)
```

Output :

```
[0. 0. 0. 0. 1. 0. 0. 0. 0. 0. 0. 0.]
```

```
original [0.91151345 0.10264149 0.65222816 0.35308932 0.77041113 0.57515432  
0.78788519 0.7291364 0.30766237 0.22994453]
```

```
sorted [0.10264149 0.22994453 0.30766237 0.35308932 0.57515432 0.65222816  
0.7291364 0.77041113 0.78788519 0.91151345]
```

```
[1. 1.11111111 1.22222222 1.33333333 1.44444444 1.55555556
```

```
1.66666667 1.77777778 1.88888889 2. ]
```

Problem #2

```
#import numpy
import numpy as np
#make og array
A = np.random.randint(1, 51, size=(3,4))
print("Original array A:\n", A)
#change last row
A[-1] = [1, 2, 3, 4]
print("\nAfter modifying last row:\n", A)
#sum all values
total_sum = np.sum(A)
print("\nSum of all elements:", total_sum)
#find min and max
min_val = np.min(A)
max_val = np.max(A)
print("Minimum value:", min_val)
print("Maximum value:", max_val)
#remove first and last row
A_reduced = np.delete(A, 0, axis=0)
A_reduced = np.delete(A_reduced, 0, axis=1)
print("\nAfter deleting first row and column:\n", A_reduced)
#remove first row first column entry
print("Not possible, numpy python arrays have a fixed size")
```

Output :

Original array A:

```
[[ 2  6  7 16]
 [22 40 17 24]
 [48 39 49 41]]
```

After modifying last row:

```
[[ 2  6  7 16]
 [22 40 17 24]
 [ 1  2  3  4]]
```

Sum of all elements: 144

Minimum value: 1

Maximum value: 40

After deleting first row and column:

```
[[40 17 24]
 [ 2  3  4]]
```

Not possible, numpy python arrays have a fixed size

Problem #3

```
def count_spaces(s): # define function that takes a string as input
    if s.count(' ') % 2 == 0: # count spaces and check if even
        print("Even number") # print if even
    else: # otherwise
        print("Odd number") # print if odd
my_string = input("Enter a string: ") # get string input from user
count_spaces(my_string) # call the function with the input string
```

Output:

```
%runfile C:/Users/noaht/untitled0.py --wdir
Enter a string: AERE161
Even number
```

```
%runfile C:/Users/noaht/untitled0.py --wdir
Enter a string: AERE161 is awesome
```

Even number

```
%runfile C:/Users/noaht/untitled0.py --wdir
Enter a string: AERE161 is so fun
Odd number
```

Problem #4

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-2, 4, 100)

y = x**3 - 26*x + 10
dy = 3*x**2 - 26
d2y = 6*x

plt.plot(x, y, label='y')
plt.plot(x, dy, label="dy")
plt.plot(x, d2y, label="d2y")

plt.legend()
plt.grid()
plt.show()
```

Output :

